

Data Management in the Cloud: What the Tech Giants are Offering



Data generated by applications is increasing by leaps and bounds almost every moment. Organisations need cloud service providers to help them manage and store this data, as well as make it available whenever they need it. Data management in the cloud has evolved into a science, and Google, Microsoft and Amazon have good solutions on offer.

Data is a collection of representation of facts as text (news), numbers (figures), graphics (datasheet), images, sound or video. Information is data in context. Without a context, data is absolutely meaningless or clueless to interpret; we create meaningful information by interpreting the context around data collected from different sources. The resulting information helps to take decisions. So, data architecture (DA) is an integral component for defining the information architecture (IA); it is concerned with the design and intent to use data in a given structured format (like databases or data sources and file systems). A data architect models the data in stages (conceptual, logical and physical), and must relate the data to each process that consumes (uses) it.

IA is the starting point for data modelling, design, and development to support business process needs. Any new models or changes to existing models must be consistent with the definitions of existing information objects. IA imposes coherence and consistency in both the definition and use of common data objects. Data management involves data architecture, information architecture, data management and governance, data security and data life cycle management.

Modern data analytics and management require high resources like CPU, memory and storage, and it is a costly investment to handle this on-premises. A cloud platform easily enables agility, scalability, reliability, availability and security for data management solutions.

Data analytics in the sports industry

We know that technology plays a vital role in modern sports in multiple ways like training, planning the game strategy, improving user experience, automating complex and humanly inefficient activities (e.g., fourth umpire reviews in cricket) and, above all, generating historical data for future analysis and training.

Today, the sports industry is using data analytics in various ways.

Dynamic modelling handles changing conditions and prepares focused data models for quicker action including match analysis, player analysis, venue analysis and forecasting. This kind of dynamic modelling helps to build robust and reliable data services quickly.